

Wireless Communication (EX751 / EX715)

Detailed PYQ

July 22, 2026

Notes

- PYQs from TU, IOE, BEX (EX751) and BEI (EX715) programmes combined (2070–2081 BS).
- BEX questions are in normal font; BEI questions are in **this monospace style**.
- Regular exam = **bold**, Back exam = regular font.
- So: **72 Ash** = BEX Regular, 73 Ma = BEX Back, **79 Bh** = BEI Regular, 80 Ba = BEI Back.
- Months: Bh = Bhadra, Ma = Magh, Ash = Ashwin, Ba = Baishakh.
- Keywords: (I) = Importance/Need/Significance, (+) = Advantage, (-) = Disadvantage, +Fig = Figure, +Eg = Example, Vs = Compare/Differentiate.

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1 Introduction

(2 Hours)

1.1 Evolution of Wireless Communications (1G–5G)

1. Briefly explain evolution of wireless communications 1G–3G [6] (70 Ma)
|→ Evolution of wireless comm. in terms of technology & worldwide market penetration [6] (70 Magh)
|→ Evolution of cellular radio 1G–3G [4] (71 Bh)
|→ Development of mobile comm., evolution path up to 3G technology [4] (74 Ma)
|→ Evolution of different generations of cellular systems [4] (**75 Bh**)
|→ Evolution from 1G to 2G, 2.5G, TDMA-based [4] (71 Ma)
2. 2G Vs 3G with +Eg of technologies; prioritized handoff + cell dragging [4+2] (**72 Ash**)
3. 3G Vs 4G mobile comm. standards Vs, technology advancement [4] (**75 Bh**) [6] (**74 Bh**)
|→ 1G, 2G, 3G, 4G Vs, technology advancement [6] (**74 Bh**)
|→ 3G Vs 4G [4] (80 Ba, 81 Ba)
|→ 2G Vs 3G Vs 4G standards, technology advancement [4] (**80 Bh**)
|→ Improvements in 2G, 3G, beyond 3G standards [6] (**70 Bh**)
4. Forward & reverse channel; 5G Vs 4G mobile communication [2+2] (**79 Bh**)
|→ Define forward and reverse channel [2] (**80 Bh**)

2 Cellular Mobile Communication Concepts

(4 Hours)

2.1 Frequency Reuse and Channel Assignment

1. Frequency reuse concept; co-channel reuse ratio [6] (72 Ma)
|→ Minimizing reuse distance maximizes spectral efficiency (I) [4] (**74 Bh**)
2. Prove co-channel reuse ratio $Q = \sqrt{3N}$; find N for omni, 120° & 60° sectoring (SIR = 15 dB, n = 4) [1+4] (71 Bh)
|→ Prove $Q = \sqrt{3N}$; find N for omni, 120°, 60° sectoring (SIR = 15 dB, n = 3) [3+5] (**79 Bh**)
|→ Derive co-channel reuse ratio $Q = \sqrt{3N}$ [4] (**80 Bh**)
|→ Co-channel Vs adjacent channel interference; prove $Q = \sqrt{3N}$; find duplex channels (N = 4, 20 MHz, 25 kHz simplex) [3+4+3] (**70 Bh**)
3. For 7-cell reuse, find min. distance between co-channel cells (area = 23 km²) [4] (**74 Bh**)

2.2 Handoff Strategies and Types

1. Handoff margin + Fig [3] (**75 Bh**)
2. Handoff strategies used in GSM [8] (70 Ma)
3. Handover process; types of handover with application [6] (74 Ma)
|→ Practical handoff considerations [4] (**79 Bh**, 73 Ma)
|→ Prioritized handoff; cell dragging [2] (**72 Ash**)

2.3 Grade of Service (GoS) and Trunking

1. GoS, measurement in blocked call cleared trunking system [3+5] (**79 Bh**)
|→ GoS + find optimal N (omni, 120°, 60° sectoring, SIR = 15 dB, n = 3) [3+5] (**79 Bh**)
2. 394 cells, 19 ch., 2% blocking, 2 calls/hr, 3 min. duration — find users & market penetration [5] (80 Ba, 81 Ba)
3. 12-cell system, 1500 km² city, find: clusters, channels/cell, max. traffic, users (2% GoS), mobiles/channel, theoretical max. users [12] (72 Ma)
4. Determine cluster size, number of clusters, max. users in service area [6] (73 Ma)
5. $S/I = n\sqrt{3N}/i_0$ — derive expression [4] (74 Ma)

2.4 Coverage and Capacity Enhancement

1. Microcell zone concept (I) + explain [3] (80 Ba, 81 Ba)
2. Sectoring Vs cell splitting: best option for capacity expansion [5] (**75 Bh**)
3. Techniques for enhancing capacity & coverage in cellular network [8] (71 Ma)

3 Radio Wave Propagation in Mobile Network Environment

(12 Hours)

3.1 Free Space Propagation and Link Budget

1. Free space propagation: receiver 10 km, 50 W TX, 900 MHz, $G_T=1$, $G_R=2$, $Z=50\ \Omega$ — find: received power, E-field magnitude, power flux density, rms voltage
[8] (80 Ba, 81 Ba)
2. TX output 165 W at 325 MHz, antenna gain 12 dBi; RX 15 km away, gain 6 dBi; find power delivered to receiver (free-space, no other losses)
[6] (80 Bh)
3. Link budget: BS 10 W, 250 MHz, 20 m coax (3 dB/100 m), 9 dBi TX ant., 25 km, 4 dBi RX ant., 0.2 dB mismatch loss — power delivered to receiver
[8] (72 Ash)
4. Radio coverage range: BS at 150 W, threshold -104 dBm, path loss at 1 m = 15 dB, exponent = 4
[6] (70 Bh)
5. Mobile 5 km from BS, $\lambda/4$ monopole (gain 2.55 dBi), E-field = 10^{-3} V/m at 1 km, 900 MHz: find length & aperture; received power (2-ray, TX ht. 50 m, RX ht. 1.5 m)
[6] (71 Bh)

3.2 Large-Scale Path Loss: Propagation Mechanisms

1. Large scale Vs small scale propagation model; propagation mechanisms in mobile environment [3+5] (72 Ash)
|→ Large scale path loss Vs small scale fading [8] (72 Ma)
2. Define path loss; parameters of a mobile multipath channel used to classify types of fading [2+4] (80 Bh)
3. Derive expression for phase difference: two-ray free space propagation model [8] (71 Bh)
|→ Diffraction in radio wave propagation; derive phase difference (Fresnel Zone Geometry model) [2+6] (79 Bh)
4. Scattering; derive expression for two-ray ground reflection model [2+8] (74 Bh)
5. Path loss Vs signal fading; time dispersion fading + types [2+6] (71 Bh)

3.3 Outdoor Propagation Models

1. Outdoor propagation models (any two) [3+3] (75 Bh) [5] (73 Ma)
2. Okumura model: median path loss for T-R = 50 km, $h_{BS}=100$ m, $h_{MS}=10$ m, suburban, correction = 9 dB, 900 MHz [4+4] (74 Ma)
|→ Okumura model, suburban, $d=50$ km, $h_{te}=100$ m, $h_{re}=10$ m, EIRP 1 kW at 900 MHz; find EIRP (dBm) & received power (RX gain 10 dB, median attenuation 43 dB, area gain 9 dB) [4+2+2] (80 Bh)
|→ Hata model: correction factor + pathloss for medium city, $f_c=950$ MHz, $h_{BS}=45$ m, $d=10$ km, $h_{MS}=5$ m; free space pathloss; compare both [4+4] (79 Bh)
|→ Hata model: max. cell radius & area, BS = 50 W, -91 dBm threshold, 900 MHz, $h_{BS}=30$ m, $h_{MS}=1$ m [6] (73 Ma)
3. What is the Ericsson Multiple Breakpoint model? [2] (80 Bh)
4. 10-km wireless link feasibility, Okumura model, suburban: AP (5 dBi, 20 dBm TX, -80 dBm sens., 100 m ht.) — client (20 dBi, 15 dBm TX, -75 dBm sens., 10 m ht.) — cables: 3 dB loss, 2.4 GHz [12] (74 Bh)

5. Two-ray mobile point-to-point: path loss, 800 MHz, TX ht. = 30 m, RX ht. = 2 m, d = 10 km; Vs free-space [4+4] (70 Ma)

3.4 Indoor Propagation Models

1. Indoor propagation models (any two) [8] (**70 Bh**)

3.5 Small Scale Fading and Multipath

1. Small scale fading: types + factors influencing [2+4+4] (**75 Bh**)
 |→ Small scale fading: types + factors [4+4] (80 Ba, 81 Ba)
 |→ Small scale fading: types + factors [2+2] (73 Ma)
 |→ Small scale fading + factors (10 questions combined) [3+6] (72 Ma)
2. Parameters of mobile multipath channel [7] (71 Ma)

3.6 Doppler Spread, Coherence Bandwidth and Time

1. Doppler spread; small-scale fading types based on Doppler spread; mean excess delay, rms delay spread, 90% & 50% coherence BW from multipath profile [4+4] (70 Ma)
2. Coherence bandwidth + coherence time (I) in mobile radio propagation [4] (73 Ma)
3. Fading classification: RMS delay spread Vs coherence time [4+4] (74 Ma)
4. Determine smallest symbol period T_s (greatest symbol rate) without equalizer from power-delay profile: BER OK when $\sigma_\tau/T_s \leq 0.1$ [8] (**75 Bh**)
5. Rayleigh Vs Rician fading channel (with expressions) [2] (71 Ma)
 |→ Rayleigh & Ricean fading distribution [5] (70 Ma, 72 Ash)
6. RMS delay spread + 90% coherence BW from power-delay profile; flat Vs frequency-selective for AMPS (30 kHz) & GSM (200 kHz) [2+6] (71 Ma)

4 Modulation-Demodulation Methods in Mobile Communications

(4 Hours)

4.1 Digital Linear Modulation: BPSK, DPSK, QPSKs

1. QPSK transmission & detection process [6] (75 Bh)
|→ QPSK transmission & detection [8] (72 Ma)
2. OQPSK transmitter & receiver; $\pi/4$ -QPSK Vs OQPSK [5+2] (71 Bh)
3. DPSK modulation: transmitter, receiver + PN sequence (I) [4+2+2] (72 Ma)

4.2 Constant Envelope Modulation: MSK, GMSK

1. MSK as special type of continuous-phase FSK & OQPSK; OQPSK Vs MSK Vs GMSK: BW efficiency, power efficiency; suitable for GSM modulation [3+3+2] (80 Ba, 81 Ba)
2. Digital modulation (+) over analog; GMSK: TX & detection process +Fig, constellation diagram [2+6] (79 Bh)
3. MSK & GMSK modulation techniques; OFDM modulator & demodulator +Fig [8] (70 Ma)

4.3 M-ary Modulation and OFDM

1. OFDM operation +Fig (block diagram of transmitter & receiver) [8] (74 Bh, 80 Bh)
|→ OFDM: generalize modulation & demodulation technique [8] (71 Ma)
|→ OFDM principle; spread spectrum modulation techniques [4+4] (70 Bh)

4.4 Spread Spectrum: DS and FH

1. Direct Sequence & Frequency Hopped Spread Spectrum techniques [4] (73 Ma)
|→ Spread spectrum Vs various MA techniques; FDMA Vs CDMA [6+2] (72 Ma)

5 Equalization and Diversity Techniques

(4 Hours)

5.1 Equalization

1. Equalization (I) in wireless comm.; signal processing to minimize ISI [2+6] (**75 Bh**)
|→ Equalization (I); basic equalization technique [2+5] (72 Ma)
|→ Fundamentals of equalization w.r.t. comm. system [4] (73 Ma)
2. MSE algorithm: optimal weights, derivation +Fig [8] (**79 Bh**)
3. Maximum Likelihood Sequence Estimation (MLSE) +Fig; time diversity (definition + two implementations) [6+1+5] (**72 Ash**)
|→ MLSE equalization +Fig [4] (71 Ma, 72 Ma)
4. Adaptive equalization algorithms (any two) [4] (73 Ma)

5.2 Diversity Techniques

1. Equalization & diversity (I); space diversity reception methods; explain any one [3+3+2] (80 Ba, 81 Ba)
2. Why implement diversity? Maximum ratio combining (space diversity) [2+2] (**80 Bh**)
3. Diversity (I) in wireless comm.; types of diversity techniques [2+6] (71 Bh)
|→ Diversity techniques; combining methods for space diversity [4+4] (74 Ma)
|→ Diversity (I); diversity combining techniques [4+6] (**70 Bh**)
|→ Various space diversity techniques +Fig [4] (73 Ma)
4. Antenna diversity techniques Vs; discuss & compare [4] (71 Ma)

5.3 RAKE Receiver

1. RAKE receiver: M-branch working [8] (**72 Ash**)
|→ RAKE receiver +Fig [4] (70 Ma)
|→ RAKE receiver [4] (79 Bh)

6 Speech and Channel Coding Fundamentals

(4 Hours)

6.1 Speech Characteristics and Vocoders

1. Characteristics of speech signal [8] (**72 Ash**) [3] (73 Ma, 79 Bh)
|→ Characteristics of speech signal; used in designing coders [8] (**72 Ash**)
2. Vocoder (I); any two predictive coders [2+6] (**70 Bh**)
|→ Formant vocoder operation; characteristics of speech signal [4+4] (71 Bh)
|→ Waveform & voice coding techniques; characteristics of speech; GSM CODEC + Fig [4+2+2] (74 Ma)
|→ Vocoders + block diagram; types of vocoders [2+3] (73 Ma)
|→ Speech coding (I); VOCODER basic concept [4+4] (70 Ma)
|→ Source coders (any two) in speech coding [6] (**74 Bh**)
3. Characteristics of speech signal; Linear Predictive Coder (LPC) + block diagram [2+6] (80 Ba, 81 Ba)
|→ LPC operation [3+5] (**79 Bh**) [2+6] (72 Ma) [3+5] (73 Ma)
4. Sub-band coding with transmitter and receiver [6] (**80 Bh**)

6.2 Channel Coding: Convolutional, Turbo, Interleaving

1. Channel coding; types of linear predictive coder [2+6] (71 Ma)
2. Convolutional encoding and decoding [5] (**74 Bh**)
3. Viterbi decoding algorithm [3] (**75 Bh**) [3] (70 Ma) [3] (**72 Ash**)
4. Turbo coding [5] (74 Ma)
5. Why is interleaving needed in wireless communication? Explain its working [4] (**80 Bh**)

7 Multiple Access in Wireless Communications

(9 Hours)

7.1 FDMA

1. FDMA non-linear effect; 25 MHz total, 100 kHz guard, 200 kHz ch. BW — find number of channels
[3+2] (80 Ba, 81 Ba)
2. FDMA (+) (-) [4] (73 Ma)
|→FDMA Vs CDMA [2+4] (70 Ma, 72 Ma)

7.2 TDMA

1. TDMA (+) over FDMA [4] (71 Ma, 73 Ma)
2. GSM time slot efficiency: 6 trailing + 8.25 guard + 26 training + 2×58 data bits; find frame efficiency
[4] (70 Bh)

7.3 Spread Spectrum Multiple Access: CDMA and FHMA

1. CDMA: characteristics, (+), (-) [2+6] (79 Bh)
|→CDMA design considerations [4] (80 Ba, 81 Ba)
|→CDMA (+) (-) [4] (73 Ma)
2. Construct a Hadamard matrix for H_8 (Walsh codes for CDMA) [4] (80 Bh)
3. SSMA: DS & FH spread spectrum techniques; (+) (-) of FDMA, TDMA, CDMA [4+4] (73 Ma)
|→Spread spectrum MA techniques; FDMA Vs CDMA [6+2] (72 Ma)
|→Spread spectrum MA types [8] (74 Ma)
4. FHMA: working with +Eg [4] (79 Bh)
|→FHMA technique with block diagram (transmitter & receiver) [6] (80 Bh) [7] (80 Ba, 81 Ba)
|→FHMA principle; hybrid spectrum MA technique (near-far mitigation) [4+6] (74 Bh)
|→FHMA [5] (75 Bh) [4] (73 Ma)
5. Explain any two hybrid multiple access techniques [4] (80 Bh)
|→Self-jamming in CDMA; FHMA +Fig; two hybrid SS MA (+) (-) [2+4+6] (72 Ash)
|→FHMA principle; near-far effect in CDMA + solution [3+4] (71 Ma)
|→Near-far effect; one hybrid SS MA technique [2+2] (71 Ma)

7.4 SDMA

1. SDMA; two hybrid SS MA techniques (near-far mitigation) [2+6] (75 Bh)

7.5 Multiple Access Comparison

1. Multiple access (define); TDMA, CDMA, SDMA [2+6] (71 Bh)
2. FDMA Vs CDMA Vs TDMA: (+) (-) each [4] (73 Ma)

7.6 Wireless LAN Standards

1. WLAN [3] (72 Ash)
2. WiMAX [3] (72 Ash) [3] (70 Ma)

8 Wireless Systems and Standards

(6 Hours)

8.1 GSM: Architecture and Channels

1. GSM architecture: operation of each component [8] (80 Ba, 81 Ba)
|→GSM architecture: draw & explain [8] (70 Ma, **79 Bh**, 73 Ma)
2. GSM traffic & control channels [8] (**75 Bh**, **72 Ash**)
3. Broadcast Control Channel (BCH) & Common Control Channel (CCCH) in GSM [4+4] (**80 Bh**)
4. GSM frame structure; TDMA frame efficiency cannot reach 100% [6+2] (74 Ma)
|→GSM frame structure; traffic & control channels during a call [4+4] (**70 Bh**)
5. Signal processing in GSM: convert speech to radio signal and back [8] (71 Bh)
6. GSM & CDMA standards; GSM architecture [4+4] (71 Ma)

8.2 CDMA Standards: IS-95 Channels

1. Channelization code; forward channels in cdma IS-95 [4] (73 Ma)
|→Draw the forward CDMA (IS-95) channel [4] (**80 Bh**)
2. CDMA Vs LTE system architecture; functions of entities [3+5] (74 Ma)

8.3 Regulatory Issues and Interference

1. Regulatory issues in wireless communication [4] (80 Ba, 81 Ba) [5] (**74 Bh**)
|→Spectrum regulations [4] (79 Bh)
2. Interference in wireless/mobile communication [4] (80 Ba, 81 Ba)

8.4 Advanced Standards: LTE, WiMAX, UMTS

1. WiMAX [3] (72 Ash, 70 Ma)
2. LTE [3] (**72 Ash**)